

Glossary

One page. Two vocabularies. Most people in the room know one of them.

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Working with an AI assistant

Agent — a program that uses a language model to take actions on your computer: reading files, writing files, running commands. Claude Code is an agent. A chat window is not.

Context — everything the model can currently see: the files it has read and the conversation so far. It cannot see files it has not opened. Most poor answers come from missing context rather than from a poor question.

Token — the unit a model reads and writes text in, roughly three quarters of a word. Context limits are counted in tokens.

Prompt — what you send to the model.

Plan mode — a state in which Claude describes what it intends to do without changing files. `Shift+Tab` enters it. Use it whenever the next step matters.

Fabrication — a confident, fluent, false statement. Sometimes called a hallucination. It is not a malfunction and it does not look different from a correct answer.

Non-determinism — the same prompt can produce different code and different results on different runs. This is why analyses are saved as scripts and rerun, not regenerated by prompting.

Reproducibility — running the analysis again gives the same answer. A saved script has this property; a prompt does not. Once a number has left your hands, in a paper, a slide or an email, recover it by rerunning the script and never by prompting a second time. See `docs/two-runs-compared.md`: two sessions, the same prompt, two different statistical tests.

CLAUDE.md — a file at the top of a project that the agent reads automatically. It is where you record conventions and constraints so that you do not have to repeat them.

RNA sequencing

Count matrix — a table of genes by samples. Each entry is the number of sequencing reads assigned to that gene in that sample.

Library size — the total number of counts in one sample. Samples are almost never sequenced to the same depth.

CPM, counts per million — counts divided by the library size, times a million. The simplest normalization. It is affected by a few very highly expressed genes.

Median of ratios — the normalization used by DESeq2. It compares each sample with a reference built from all samples and takes a median, so a small number of dominant genes does not distort it.

log2 fold change — the difference between two conditions on a log2 scale. 1 means twice as high, 2 means four times as high, minus 1 means half.

p-value — the probability of seeing a difference at least this large if there were no real difference.

Multiple testing correction — an adjustment applied because thousands of genes are tested at once. Testing 4000 genes at a threshold of 0.05 gives about 200 significant results even when nothing is real.

FDR, false discovery rate — the expected proportion of your significant results that are false. The Benjamini-Hochberg procedure controls it. The corrected value is usually written `padj`.

Batch — a group of samples processed together. Samples in the same batch resemble each other for reasons unrelated to the biology.

Gene set enrichment — a test of whether your significant genes include more members of a defined set than chance would give. It is a statement about a list, not about a biological process.